**MIL-STD-1226A**
**29 October 1965**

**SUPERSEDING**
**MIL-STD-1226**
**3 October 1961**

# MILITARY STANDARD

# ENGINE, GASOLINE: AIR-COOLED, 1-1/2 BHP, 4-CYCLE, MILITARY DESIGN, MODEL 1A08 INSTALLATION PROCEDURES




FSC 2805

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# DEPARTMENT OF DEFENSE

## WASHINGTON, D. C. 20301

Engine, Gasoline, Air-Cooled, 1-1½ BHP, Military Design, Model 1A08, Installation Procedures.

MIL-STD-1226A

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommanded corrections, additions, or deletions should be addressed to the Director, U. S. Army Mobility Equipment Center, Research, Development, and Engineering Directorate, ATTN: OMEFB-RDE-KM, Fort Belvoir, Va., 22060.



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## 1. SCOPE

**1.1 Coverage.** This standard covers the recommended installation procedures for the model 1A08, 1½-hp, Military design engine.

**1.2 Objectives.** The objective of this standard is to insure compatibility of the engine and the end item of equipment.

## 2. REFERENCED DOCUMENTS

**2.1 Governmental.** The issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

MIL-V-173 — Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).

MIL-T-704 — Treatment and Painting of Material.

MIL-V-1137 — Varnish, Electrical-Insulating (for Electro-Motive Equipment).

MIL-B-11040 — Belt, V; Engine Accessory Drive.

MIL-E-55301 — Electromagnetic Compatibility.

STANDARDS

MS51064 — Pulley, Groove: Engine Accessory Drive Belts.

MS51065 — Belts, V; Engine Accessory Drive (0.380 Inch Nominal Width).

MS51066 — Belts, V; Engine Accessory Drive (0.500 Inch Nominal Width).

MS51067 — Belts, V; Engine Accessory Drive (11/16 Inch Nominal Width).

MS51068 — Belts, V; Engine Accessory Drive (3/4 Inch Nominal Width).

MS51069 — Belts, V; Engine Accessory Drive (7/8 Inch Nominal Width).

MS51070 — Belts, V; Engine Accessory Drive 1 Inch Nominal Width).

MS51086 — Filter, Fluid, Pressure and Strainer, Sediment; Automotive, Fuel (10 GPH, Coarse Filtration).

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

**2.2 Non-Governmental.** The issues of the following documents in effect on date of invitation for bids form a part of this standard to the extent specified herein.

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook



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(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N.Y. 10017.)

## 3. DEFINITIONS

**3.1 Definitions.** For the purpose of this standard, the following definitions shall apply.

3.1.1 *The Military Design engine Model 1A08.* A single-cylinder air-cooled, overhead-valve, 8-cubic-inch-displacement engine having a rating of 1½ net continuous horsepower rating at 3,600 rpm.

3.1.2 *Maximum net corrected brake horsepower.* The maximum net corrected brake horsepower rating with all accessories (including fan, muffler, and air cleaner) at any engine speed within the operating range is the maximum observed horsepower available from the engine at wide-open throttle corrected to standard atmospheric conditions.

3.1.3 *Intermittent net brake horsepower.* The intermittent net brake horsepower rating is 90 percent of the maximum net corrected brake horsepower.

3.1.4 *Continuous net brake horsepower.* The continuous net brake horsepower rating is 1.5 observed horsepower at 3,600 rpm.

## 4. GENERAL REQUIREMENTS

**4.1 Safety.** When installed in the end item, rotating or reciprocating parts and parts subject to high temperatures that are so located as to become a hazard to operating personnel and equipment shall be insulated, fully enclosed, or guarded. Exhaust mufflers and piping shall be located to minimize hazard to operating personnel.

**4.2 Use conditions.** The installation shall withstand shock loads as specified in the end item specification. The installation shall be such that the engine will not be required to operate in a tilted position of more than 15 degrees from the horizontal in any plane at any time.

**4.3 Design simplicity.** The design of the end item shall be such that complete removal of the engine from the driven component or the driven component from the engine can be accomplished with minimum disassembly and without the use of special tools.

## 5. DETAIL REQUIREMENTS

**5.1 Power requirements.** The maximum horsepower required to drive the end item, including power transmission system under the environmental extremes specified for the end item, shall not exceed the net continuous horsepower rating of the engine unless intermittent operation is indicated in the end-item specification, at which time the intermittent power requirement shall not be exceeded.

**5.2 Operational requirements.** The end item design and location of hoods and other external components shall not cause the engine to exceed the operational limits specified in table I under all operating conditions of load and environmental extremes specified in the end-item specifications.

TABLE I. *Operational temperature limits*

| | Location | Maximum temperature °F. |
|---|---|---|
| Ambient air | In the vicinity of the engine | 120 |
| Cylinder head | Under spark plug | 475 |
| Lubricating oil | In the oil sump | 250 |
| Cooling air | At cooling air outlet | 225 |
| Carburetor inlet air | Air cleaner inlet | 150 |



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5.3 **Maintainability.** The engine installation shall permit maximum accessibility for replacement, adjustment, and repair to all parts of the engine with minimum disturbance to adjacent parts of the end item without the use of special tools.

5.3.1 *Cooling and starting system.* Clearance shall be provided for:

(a) Winding the starter rope on the starter pulley with a gloved (arctic type) hand without interference.

(b) Removal of the flywheel housing and flywheel without removing other components of the end item.

5.3.2 *Fuel system.* Clearance shall be provided for:

(a) Carburetor adjustments.

(b) Removal of the fuel filter.

5.3.3 *Induction system.* Clearance shall be provided for:

(a) Servicing the air cleaner.

(b) Operating the winterization air control on dry-type air cleaners.

5.3.4 *Lubricating system.* Clearance shall be provided for:

(a) Removal and insertion of the oil-gage rod and addition of oil with or without a removable spout or funnel.

(b) Removal and replacement of the oil drain plug and for draining the oil. The oil shall drain completely and shall not flow over any part of the end item. An extension to the crankcase drain system is permissible.

5.3.5 *Governor system.* Clearance shall be provided for governor speed and linkage adjustments.

5.3.6 *Ignition system.* Clearance shall be provided for removing, replacing, and adjusting spark plugs, high tension cables, breaker points, and capacitors.

5.3.7 *Valves.* Clearance shall be provided for removing rocker-arm covers and adjusting valve tappet clearance.

5.4 **Engine mounting requirements.** Mounting brackets shall be in accordance with figures 1, 2, 3, or 4 unless other means of mounting is approved by the contracting officer. The mounting brackets shall be furnished by the end-item manufacturer and shall be installed in such a manner as to permit assembly without deflection or deformation of brackets or engine mounting pads.

5.4.1 *Vibration isolators.* Suitable vibration isolators which stress an elastomer in shear or compression or both shear and compression shall be used on all installations when shock mounting is specified in the end-item specification. The method of calculation for determining the maximum allowable spring force of four vibration isolators equally spaced about the center of gravity of the end item for beam mounting (see figure 5) shall be as specified in 5.4.2.1. When vibration isolators are not equally spaced about the center of gravity a complete analysis shall be conducted to determine the maximum permissible spring force for each isolator.

5.4.2 *Design requirements for direct mounting of end item.* The maximum resultant moment that the equipment mounting pads can absorb without deformation is 650 lb-ft. This limiting moment shall not be exceeded. Cantilever mounting of end items to the engine is prohibited unless specifically detailed on Military design



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drawings. Cantilever mounting of the engine to the end item is permissible provided the applicable end-item specification does not require free fall test.

5.4.2.1 *Method for calculating maximum allowable spring force.* The final spring force shall not apply to the weight of the end item a G-magnification factor that will cause a moment exceeding 650 lb-ft, when calculated as shown in figures 6 through 9.

5.4.2.2 *Procedure.* The following procedure shall be followed in selecting four vibration mounts equally spaced about the center of gravity of the end item to withstand freefall, shock-load requirements.

(a) Determine weight of equipment; e.g., $W_{eq} = 80$ lb.

(b) Select a vibration isolator, and compute spring rate; e.g., k = 3,000 lb/inch (obtain spring force from manufacturer).

(c) Calculate the static deflection by the following formula:

$$\delta o = \frac{W_e + W_{eq}}{4k}$$

Where, $\delta o$ = static deflection (inches)

k = spring rate in lb/inch = 3,000 lb/inch

$W_e$ = weight of engine = 27 lb

$W_{eq}$ = weight of equipment = 80 lb

(d) from design chart (see figure 6) determine final spring force $F_s = 1{,}700$ lb.

(e) Determine factor e (see figure 7) which is a function of L = distance of CG of equipment from engine equipment mounting pads, and D = distance of equipment support from engine equipment mounting pads.

Example, L = 8 inches

D = 14 inches

Therefore, factor e = 2.5

(f) Determine maximum permissible force (see figure 8).

For e = 2.5

$W_{eq} = 80$ lb

Therefore, Maximum permissible spring force = 5,100 lb

*Note.* This value is higher than the final spring force and the four shock mounts are suitable if the natural frequency and total deflection of the mounts are within the design specification.

(g) Determine natural frequency and total deflection (see figure 9).

Static deflection $\delta o = 0.0089$ inch

Natural frequency F = 2,000 CPM

Total deflection t = 0.57 inch

F shall not be greater than 70 percent of the engine governed speed.

**5.5 Power transmission requirements.** The drive unit shall be connected to the engine drive shaft by one of the following methods. Torsional vibration determinations shall be made as specified in 5.5.5.

5.5.1 *Rigid-quill coupling.* When a rigid-quill coupling is used on the driven equipment shaft the coupling shall be in accordance with figure 10.



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5.5.2 *Flexible coupling.* When a flexible coupling is used between the engine and end item, the coupling shall provide the correct torsional flexibility and a sufficient degree of dampening to insure that the maximum vibration torque does not exceed 410 lb-in (see 5.5.5.2).

5.5.2.1 *Shaft misalinement.* The misalinement of the engine shaft and driven equipment shaft shall not exceed the limits shown in figure 11 when a flexible coupling is used between the engine and end item.

5.5.2.2 *Coupling installation.* The coupling shall be installed on the engine shaft by a taper connection in accordance with figure 12.

5.5.3 *Belt drive.* V-belts used shall conform to MIL–B–11040. The belt size and pulley dimension shall conform to MS51064 through MS51070, as applicable.

5.5.3.1 *Pulley alinement.* The alinement of pulleys shall insure optimum belt life and power transmission.

5.5.4 *Gear drive.* When an independent reduction gear drive is used, it shall be coupled to the engine through a suitable flexible coupling (see 5.5.2) or V-belt drive (see 5.5.3).

5.5.4.1 *Direct-mounted gear-reduction drive.* When a direct-mounted gear-reduction drive is used, it shall be coupled to the engine power-takeoff shaft as specified in 5.5.5.1.

5.5.5 *Method for calculating torsional vibration stresses.* The maximum allowable vibration torque shall not exceed 410 lb-in. for the shaft between the reciprocating masses of the engine and the attachment point of the driven equipment.

5.5.5.1 *Procedure for directly connected end item.* The following procedure for torsional vibration calculation shall be followed:

(a) Conditions. The characteristics of the directly-connected end item are:

End item inertia $J_M = 0.15$ lb-in-sec$^2$
or $WR^2 = 57.8$ lb-in$^2$

Engine speed range 3,500 to 3,750 rpm.

(b) Evaluate the critical speeds occuring in the engine speed range of 3,500 to 3,750 rpm from figure 13.

For $J_M = 0.15$ lb-in-sec$^2$ the nearest critical speed occurs at $N_c = 3{,}440$ rpm for $n = 3\frac{1}{2}$

*Note.* The engine shall not be operated continuously at any critical speed which will result in a vibratory torque in excess of 410 lb-in.

(c) Determine the vibratory torque in the shafting from figure 14.

For $J_M = 0.15$ lb-in-sec$^2$ and $n = 3\frac{1}{2}$. The vibratory torque $T_c = 508$ lb-in. Since the critical speed $N_c = 3{,}440$ rpm is below the minimum speed range, the actual vibratory torque at $N = 3{,}500$ rpm shall be determined.

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(d) Calculate speed ratio

$$\frac{N = 3500}{N_c = 3440} = 1.017$$

(e) Determine average torque magnifier from figure 15.

For speed ratio = 1.017, the average torque magnifier

$$T/T_c = 0.670,$$

thus the vibratory torque at 3,500 rpm

$$T = 0.670 \times 508 = 341 \text{ lb-in.}$$

This value is within the vibratory torque limits for the crankshaft.

5.5.5.2 *Procedure for flexible coupling connected load.* The following information shall be determined if a flexible coupling is used:

(a) The inertia of the driven machine (see appendix 10.10)

(b) The stiffness of the coupling (furnished by the coupling manufacturer) shall not exceed the maximum permissible stiffness (see figure 16).

5.6 Miscellaneous design requirements.

5.6.1 *Alteration.* The engine shall not be altered in any manner by the end-item manufacturer for assembly or installation purposes.

5.6.2 *Fuel tank.* The capacity of the fuel tank for the end item may be determined from table III. The location of the fuel tank and maximum allowable length of fuel supply line shall not exceed the limits specified in figure 17.

5.6.3 *Cooling-air outlet.* Clearance shall be provided for the outlet of cooling air as shown on figure 18.

5.6.4 *Noise reduction.* In applications where additional noise reduction is required, a secondary muffler shall be directly connected to the muffler outlet; however the secondary muffler shall be self supporting.

Custodians:
Army — MO(ERDL)
Navy — YD
Air Force — 69

Review activities:
Army — MO GL
Navy — YD MC
Air Force — 69
DSA — CS

User activities:
Army — EL
Navy — None
Air Force — None

Preparing activity:
Army — MO(ERDL)
Project No. 2805-0115



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## 10. APPENDIX

**10.1 General Description of Engine.** The model 1A08 Military design engine is a single-vertical-cylinder, overhead-valve, air-cooled, spark-ignition, 4-cycle, gasoline engine. The net continuous brake horsepower rating is 1½ hp at 3,600 rpm. The engine will develop 85 percent of the maximum brake horsepower specified in the performance charts. The engine is treated and painted in accordance with MIL–T–704.

**10.2 Engine accessories.** The engine as supplied will be equipped with all accessories as outlined in the installation drawing (see figure 19) and shall be ready for immediate operation after the engine is correctly serviced and deprocessed. The engine has the following features.

10.2.1 *Dry-air cleaner.* The engine air cleaner is of the dry (replaceable element) type with a restriction indicator (see figure 20). Cleaning and servicing instructions are embossed in the air cleaner body.

10.2.2 *Fuel filter.* The fuel filter consists of a strainer and the sediment bowl. It is provided for use between the fuel supply and the fuel pump. The filter body is tapped for 1/8-inch NPT fuel-line fittings.

10.2.3 *Fuel pump.* The fuel pump is a single-acting, diaphragm-type pump rated for a static pressure range between 1.5 psi minimum to 5.00 psi maximum, measured at a point 16 inches above fuel pump outlet with pump cam action at 1,800 rpm. The flow pressure head is between 1.75 psi and 3.00 psi with a normal head pressure of 2 psi. The rated capacity of the pump at 1,800 strokes per minute is 5 gallons per hour.

10.2.4 *Muffler.* The engine exhaust system is comprised of a combination muffler-manifold which is completely housed within the engine shrouding. An external muffler is not required for normal installation to limit the engine noise since the muffler-manifold is capable of attenuating the overall engine noise level to 83 db when measured at a distance of 50 feet. The muffler outlet is a standard SAE two-hole flange (see figure 19) positioned at an angle of 45 degrees to enable the exhaust gases to escape into the atmosphere away from the engine.

10.2.5 *Cooling system.* A single cooling-heating system is provided to insure operation over the specified temperature range from minus 25° F. to 120° F. Cooling air is provided by a centrifugal fan integral with the diecast aluminum flywheel. The cooling air is forced over the top portion of the cylinder barrel and is discharged across the bottom of the engine oil pan. The system is designed to maintain safe engine temperatures over the entire load range. At rated speed, the airflow is 82 cubic feet/minute.

**10.3 Engine illustration and installation.** Figure 21 is a photograph of the model 1A08 Military design engine illustrating the major components viewed from the rope starter end. Figure 20 is a similar illustration of the engine viewed from the engine power takeoff shaft. The installation drawings are presented in figure 19 and illustrate the overall dimensions of the engine, the location of the major components and the detailed dimensions of the engine mounting pads, equipment mounting pads, and the power takeoff shaft. The dimensions indicated shall be used to assist in the design of the installation.

**10.4 Engine specifications.** The engine specifications are as listed in table II:



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TABLE II. *Engine specifications*

*Military design engine, model 1A08:*

A. Engine:

| | |
|---|---|
| Number of cylinders | 1 |
| Bore, inches | 2.25 |
| Stroke, inches | 2.00 |
| Total displacement, cubic inches | 8.00 |
| Rated horsepower at 3,600 rpm | 1.5 BHP |
| Maximum horsepower at 3,600 rpm | 2.5 BHP |
| Maximum torque at 3,600 rpm, lb-ft | 3.75 |
| Compression ratio | 6.0:1 |
| Speed range, rpm (for continuous operation) | 3,400–3,800 |

B. Fuel system:

| | |
|---|---|
| Fuel pump | diaphragm |
| Fuel filter | MS51086 |
| Fuel consumption at rated load and speed, lb/bhp-hr | 1.10 |

C. Lubrication system:

| | |
|---|---|
| Lubrication system | splash-vapor |
| Oil sump capacity, pints | 1.00 |
| Oil consumption, maximum at rated load and speed, lb/bhp-hr | 0.025 |

D. Ignition:

| | |
|---|---|
| Ignition system | magneto |
| Spark plug | 18-mm |
| Electromagnetic compatibility | MIL–E–55301 |

E. Governor characteristics:

| | |
|---|---|
| Speed regulation, percent | 3 |
| Rated load and speed, rpm | 3,600 |
| No-load speed, rpm (maximum) | 3,708 |
| Engine speed stability (at constant value of load), percent | plus or minus 1/3 of 1 |
| Maximum speed surging characteristics: | |
| Rated load to no load, seconds | 4 |

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TABLE II. *Engine Specifications*—(Cont'd)

| | |
|---|---|
| No load to rated load, seconds | 4 |
| Maximum speed change during 4-second surging period, percent | 6 |
| Rated load to no load, rpm | 216 |
| No load to rated load, rpm | 216 |

F. Engine and accessories:

| | |
|---|---|
| Air cleaner | dry type |
| Cranking system | rope |
| Main bearings | tapered roller |
| Crankshaft rotation, viewed from drive end | counterclockwise |
| Dry weight, pounds | 29.5 |
| Life between major overhaul or rebuild, hours | 1,500 |

G. Overall dimensions:

| | |
|---|---|
| Height, inches | 15 |
| Length, inches | 12⅜ |
| Width, inches | 14¼ |

H. Environmental extremes:

| | |
|---|---|
| Engine operation | plus 120° F. to minus 25° F. |
| Engine storage | plus 150° F. to minus 80° F. |
| Engine starting capability without preheat | plus 120° F. to minus 25° F. |
| Starting with preheat | minus 25° F. to minus 50° F. |
| Fungusproofing | MIL–V–173 and MIL–V–1137 |
| Humidity extreme, percent relative humidity | 85% at 85°F. |
| Tilt operation, degrees in any plane (maximum) | 15 |
| Maximum elevation for rated power, feet | 8,000 |

10.5 **Fuel consumption.** The part-throttle fuel consumption characteristics (at 3,600 rpm) of the 1½ bhp Military design engine are presented in the following table. There is a 10 percent increase in the fuel consumption at 3,800 rpm and a 10 percent decrease at 3,400 rpm. The effect of altitude is insignificant.



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TABLE III. *Fuel consumption*

| | | | | | |
|---|---|---|---|---|---|
| Brake horsepower | ¼ | ½ | 1.0 | 1½ | 2.0 |
| Fuel consumption, gal/hour | 0.18 | 0.20 | 0.22 | 0.25 | 0.27 |

**10.6 Moment of inertia of end item.**

**10.6.1** Determine the polar moment of inertia of a balanced pivoted mass which is not easy to remove or handle (such as the rotor of an electric motor) as follows.

(a) Support the rotating mass in a horizontal plane on antifriction bearings or on knife edges.

(b) Attach a known weight, W, to the mass at a distance, L, from the axis of rotation. (When the mass is not accessible and the shaft rotates with it, attach the weight to the shaft with a light rigid rod of length L).

(c) Set the system in oscillation at amplitudes of 10 degrees or less and measure the period T.

(d) Determine the moment of inertia using the following formula:

$$J = WL \left\{ \left( \frac{T}{2\pi} \right) - \frac{L}{G} \right\}$$

Where:

J = moment of inertia, lb-in-sec$^2$

W = weight of mass, lb

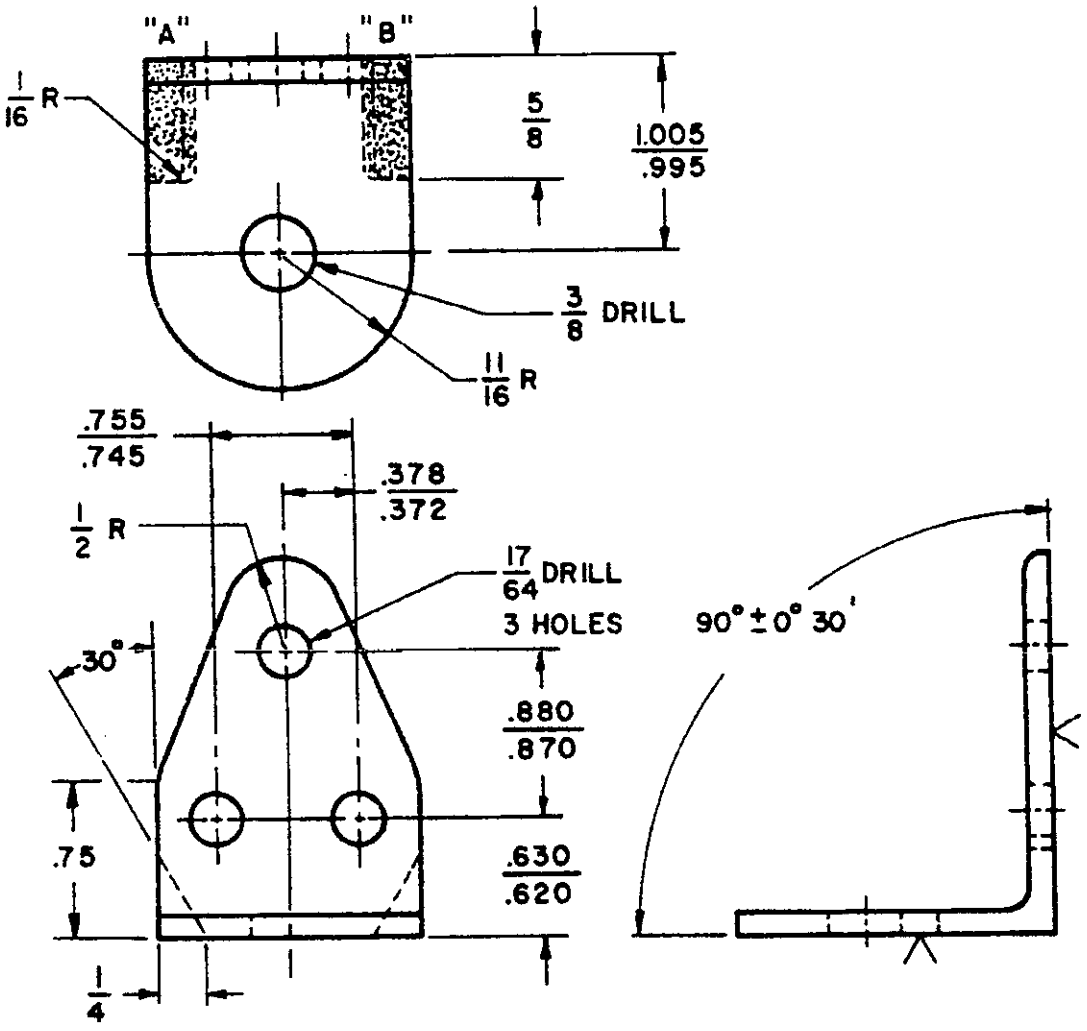
L = length of rod, inches

T = second/period

G = acceleration of gravity = 386 in/sec$^2$



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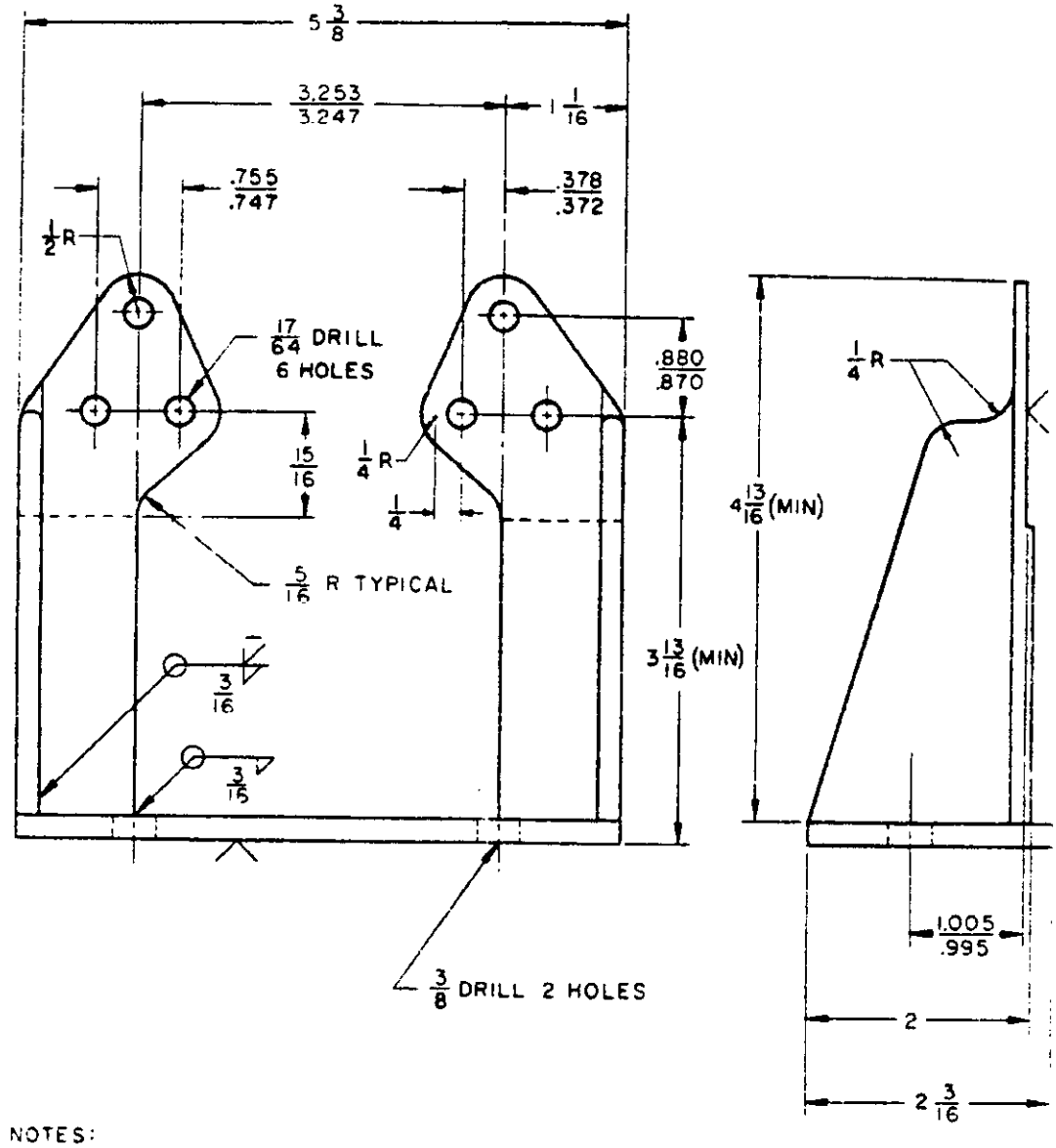
1. MATERIAL: 2 x 2 x 1/8 ANGLE, STEEL (MIN)
2. BLANK OUT SHADED AREAS "A" OR "B" FOR AIR DUCT CLEARANCE (2 ANGLES)
3. FOUR BRACKETS / ENGINE REQUIRED

FIGURE 1. *Engine mounting bracket.*

CX–423



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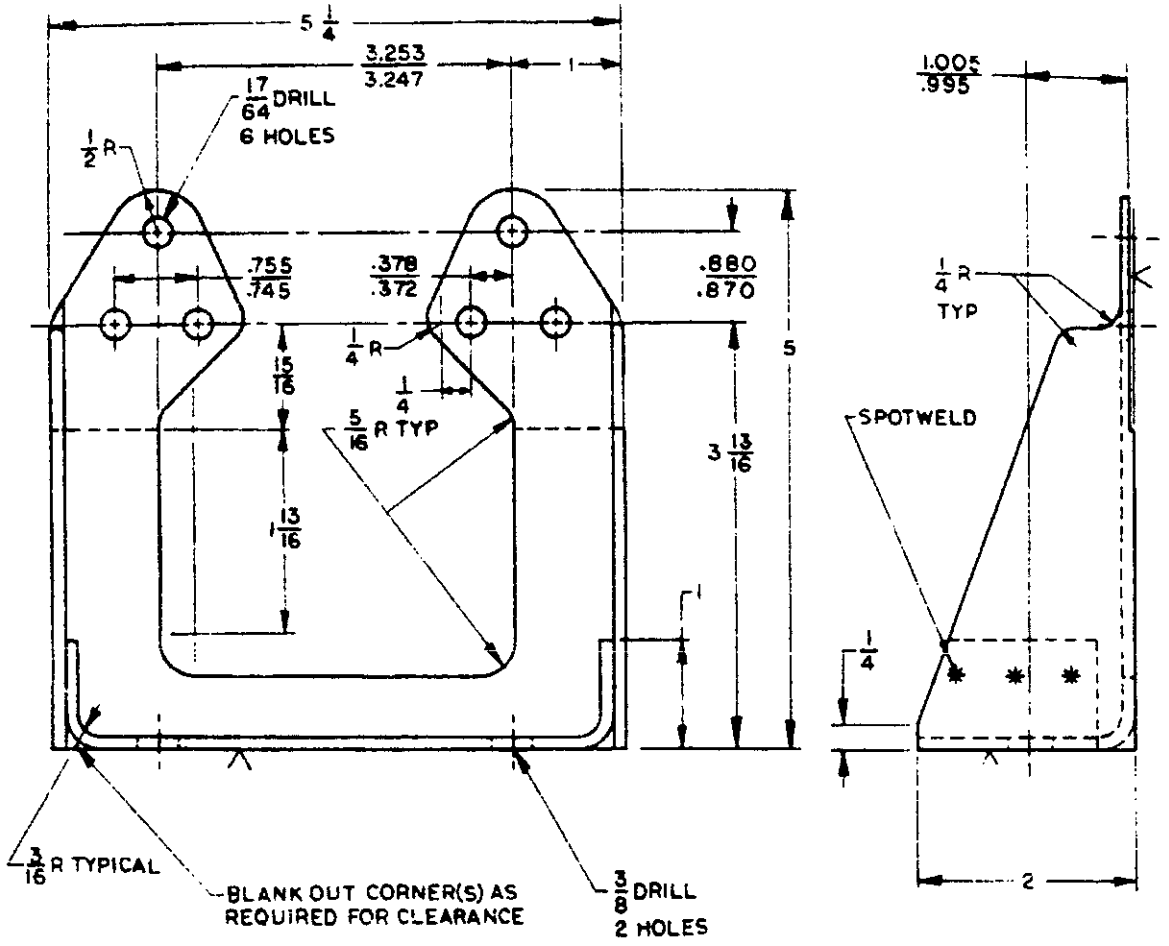
1. MATERIAL: 3/16 STEEL PLATE
2. FRACTIONAL TOLERANCE ±1/32
3. NUMBER REQUIRED: 2/ENGINE
4. "V" SURFACES MUST BE FLAT AND SQUARE WITHIN ±.007 T.I.R.

FIGURE 2. *Welded engine mounting bracket.*

CX-425



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NOTES:

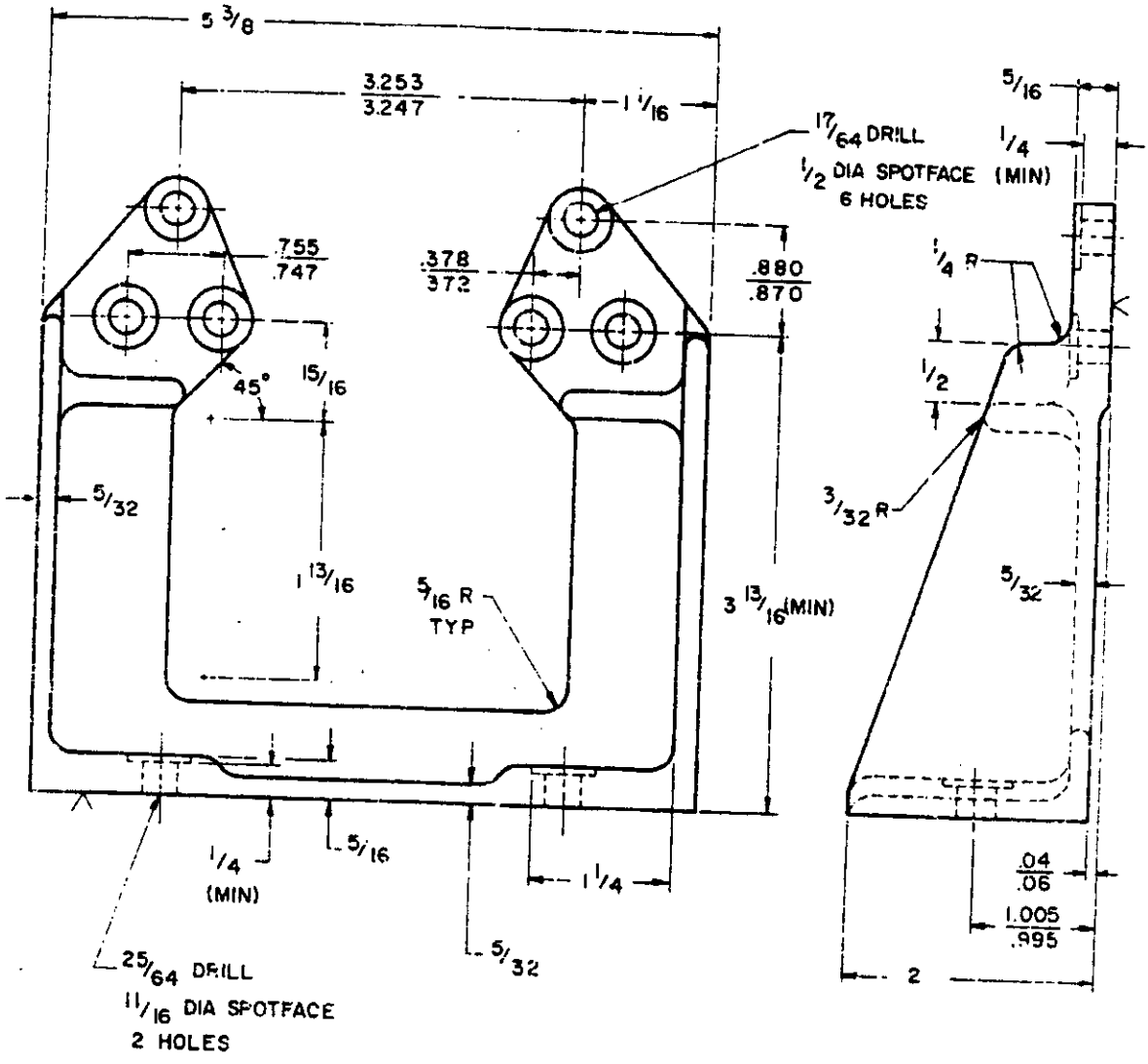
1. MATERIAL: $\frac{1}{8}$" STEEL PLATE (MIN)
2. FRACTIONAL TOLERANCES: $\pm \frac{1}{32}$"
3. NUMBER REQUIRED: 2/ENGINE
4. "V" SURFACES TO BE FLAT AND SQUARE TO EACH OTHER WITHIN ±.005" T.I.R.

*Figure 3. Stamped engine mounting bracket.*

CX-424



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NOTES:

1. MATERIAL: SAE 309 ALUMINUM
2. DRAFT ANGLES 1°–2°
3. BLEND ALL RADII TO A SMOOTH CONTOUR
4. FRACTIONAL TOL. ± 1/32"
5. FILLETS AND ROUND 3/16 R UNLESS OTHERWISE SPECIFIED
6. HOLES MAY BE CORED
7. "V" SURFACES TO BE FLAT AND SQUARE TO EACH OTHER WITHIN ± .005 T.I.R.
8. NO. REQUIRED 2 / ENGINE

FIGURE 4. *Cast engine mounting bracket.*

CX-432



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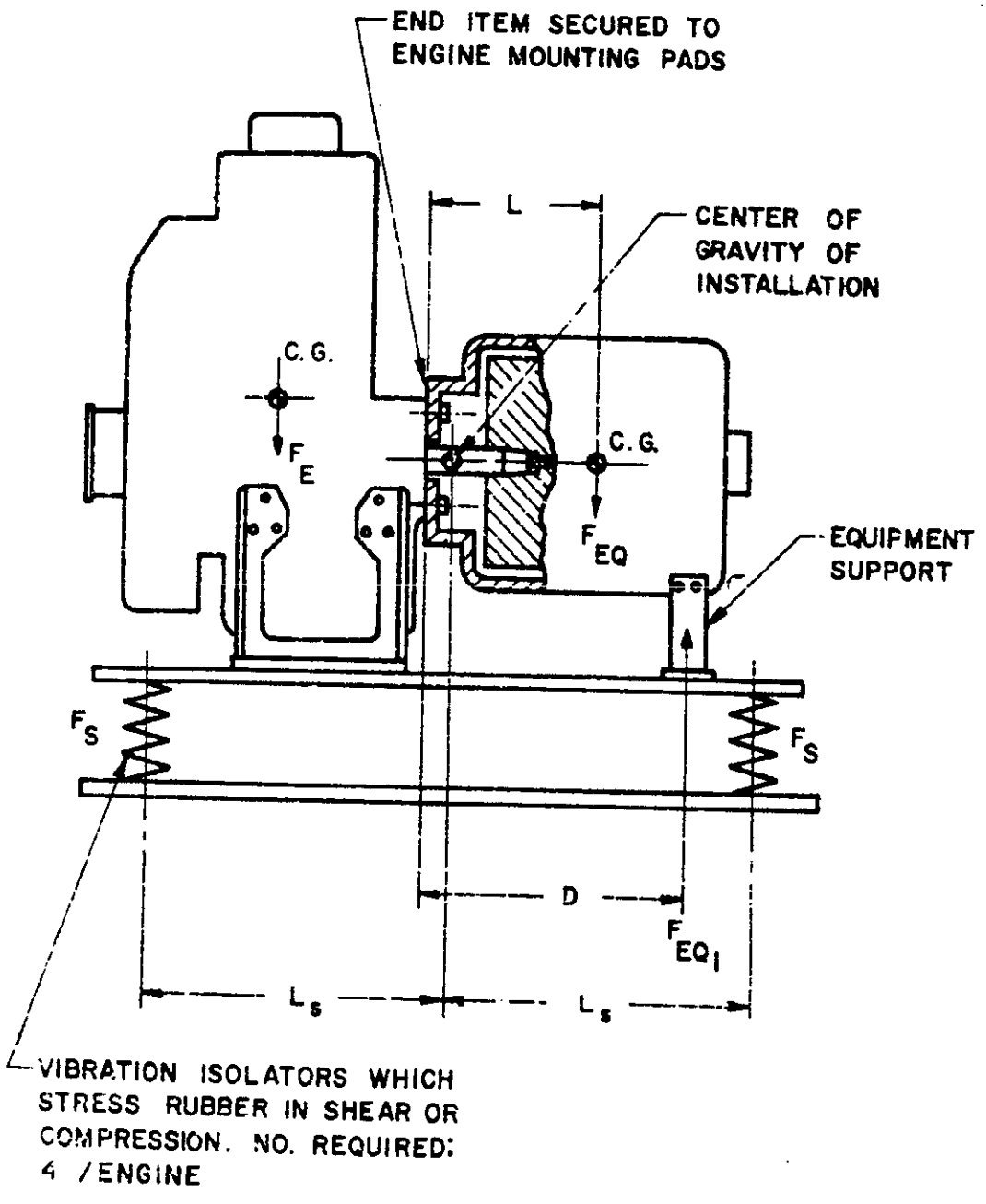



FIGURE 5. *Beam mounting installation.*

CX–426

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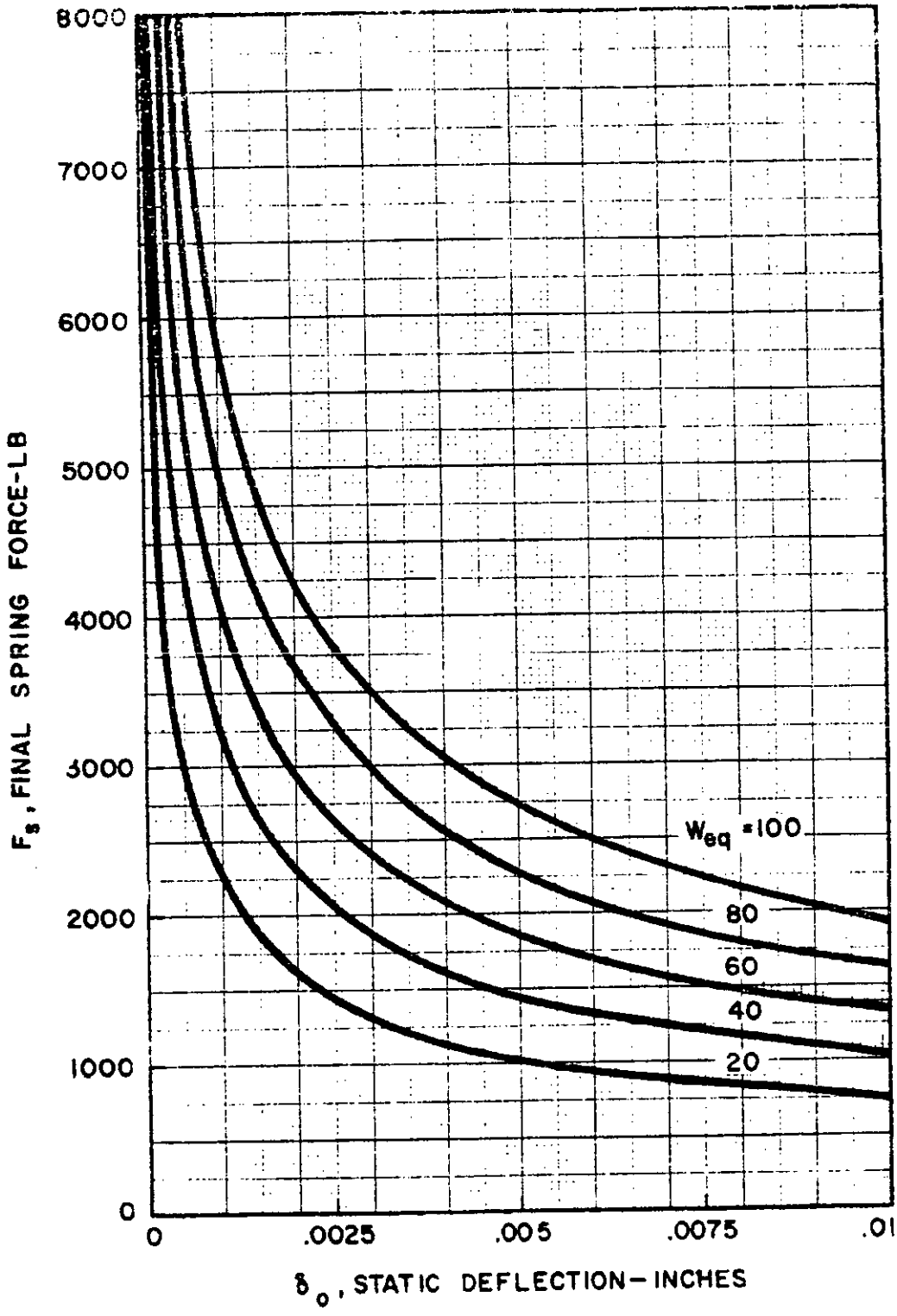



FIGURE 6. *Final spring force (per spring).*

CX–441



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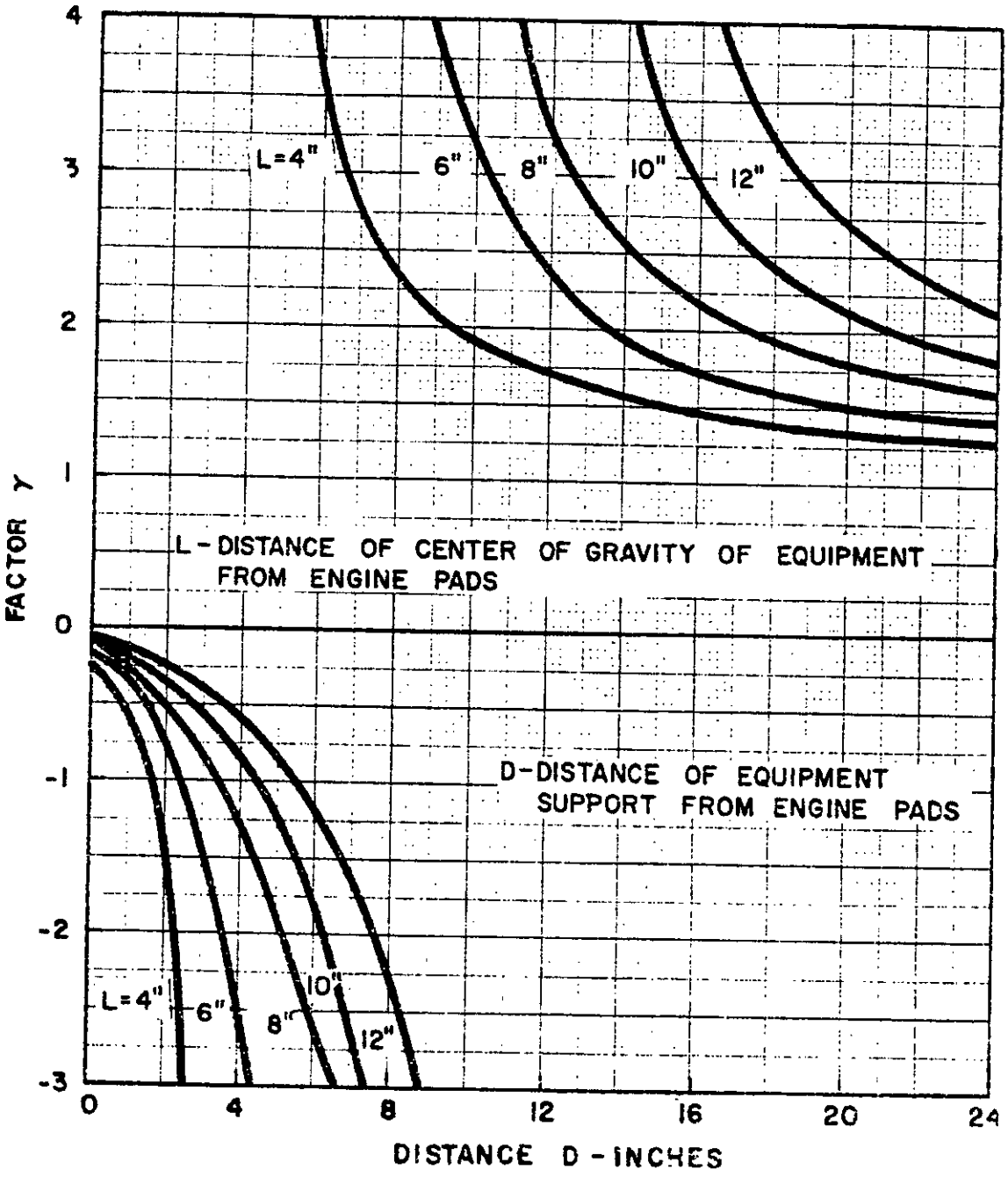



FIGURE 7. *Factor y as a function of L and D for lower mounting.*

CX–446

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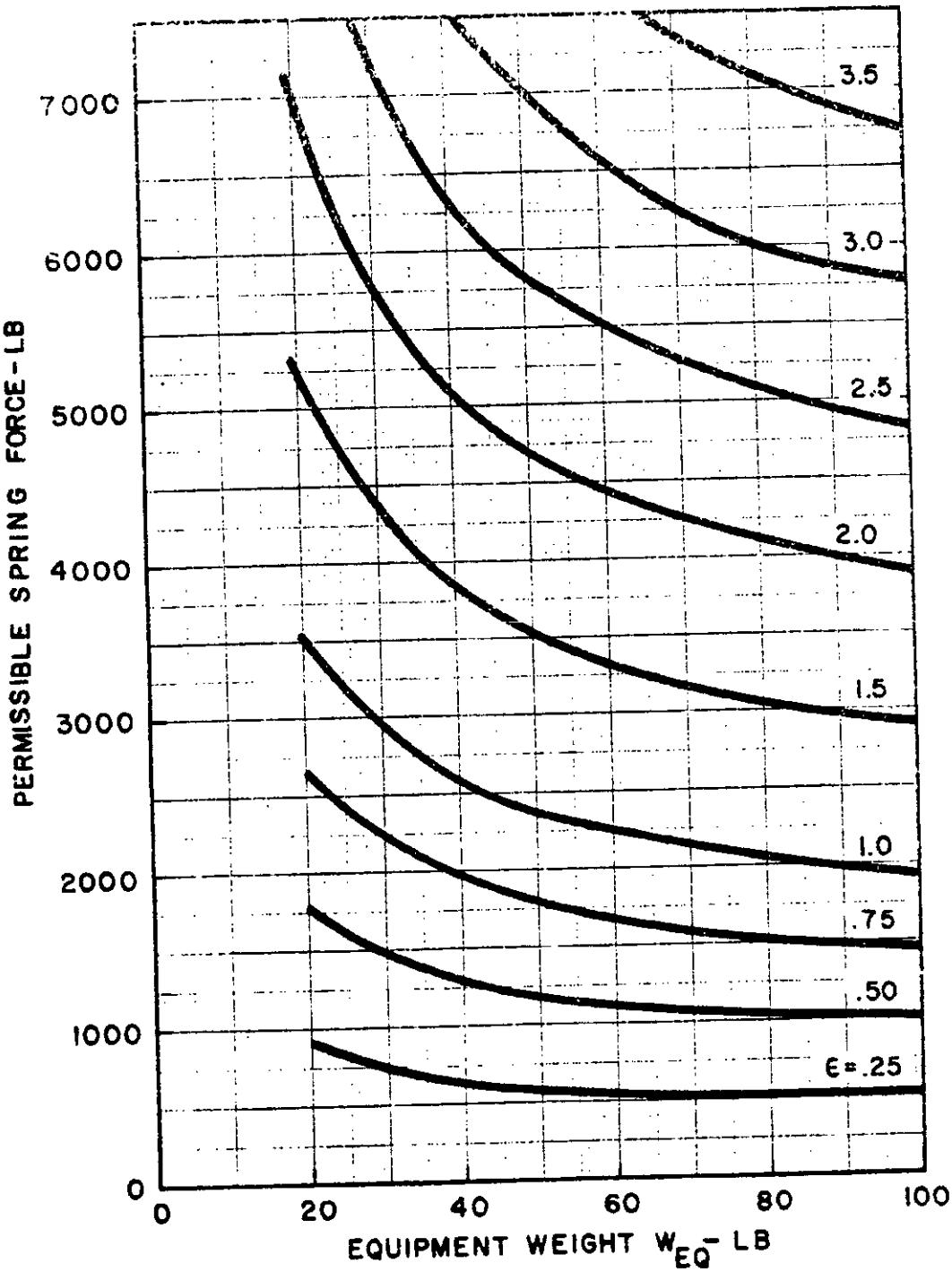



FIGURE 8. *Permissible spring force (per spring) for beam loading.*

CX-417



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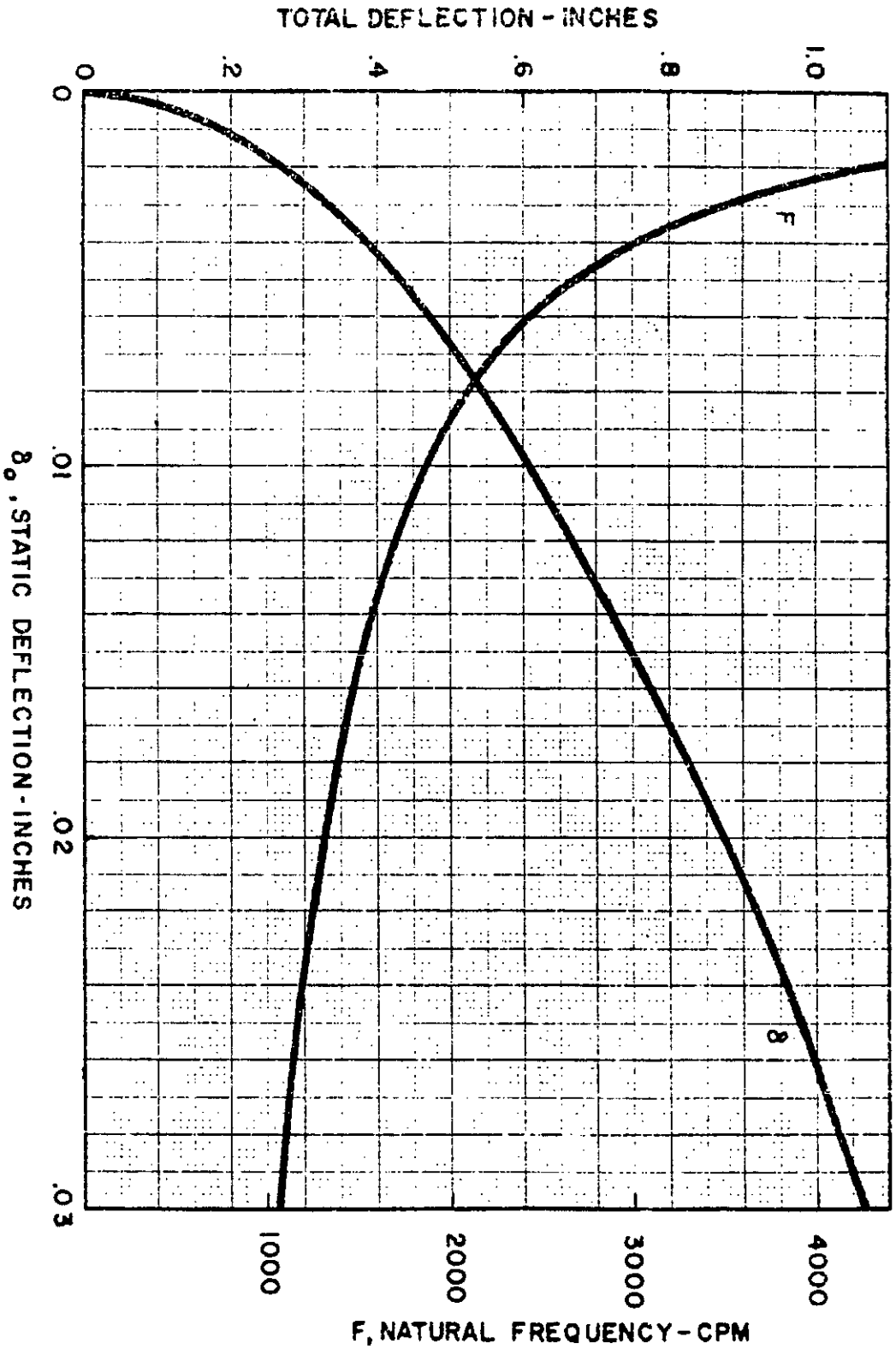



FIGURE 9. *Total deflection and natural frequency as a function of static deflection.*

CX-448

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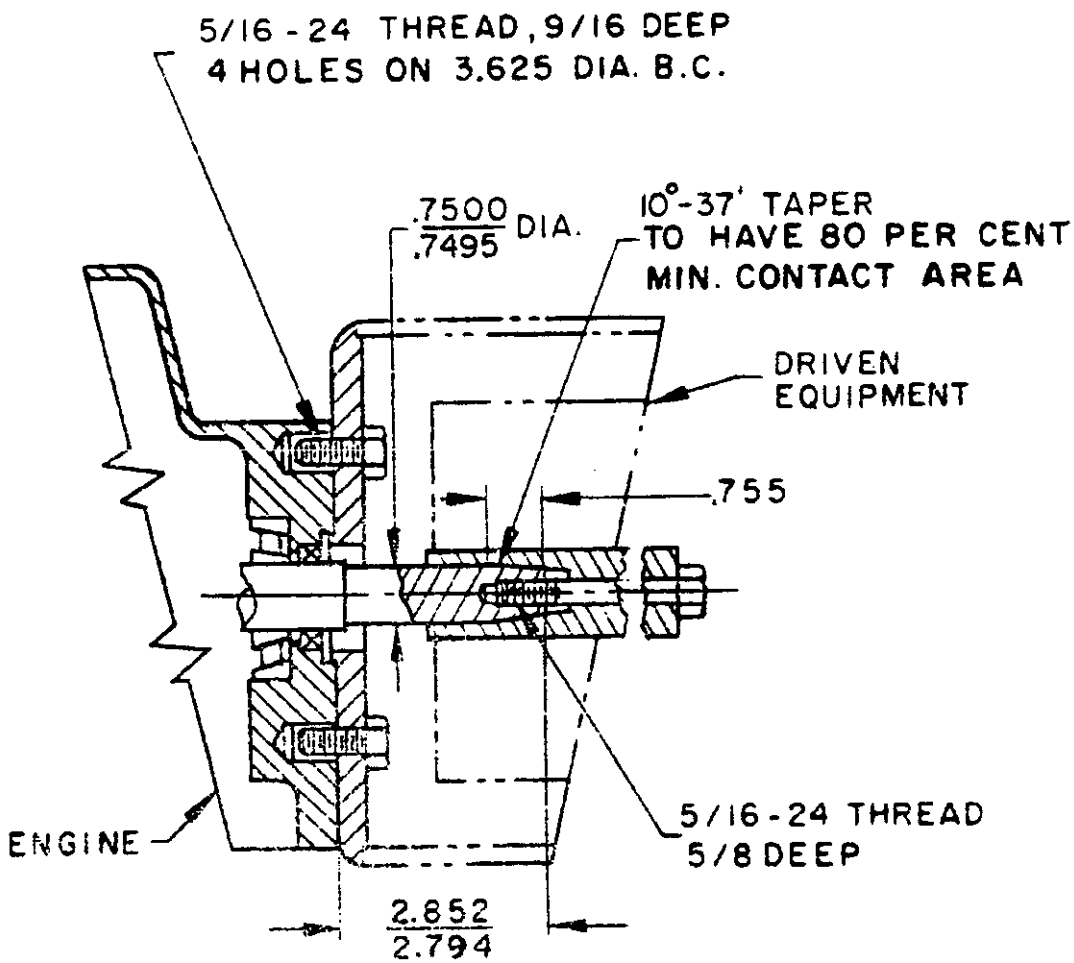



FIGURE 16. *Rigid spill-type coupling.*

CX-422



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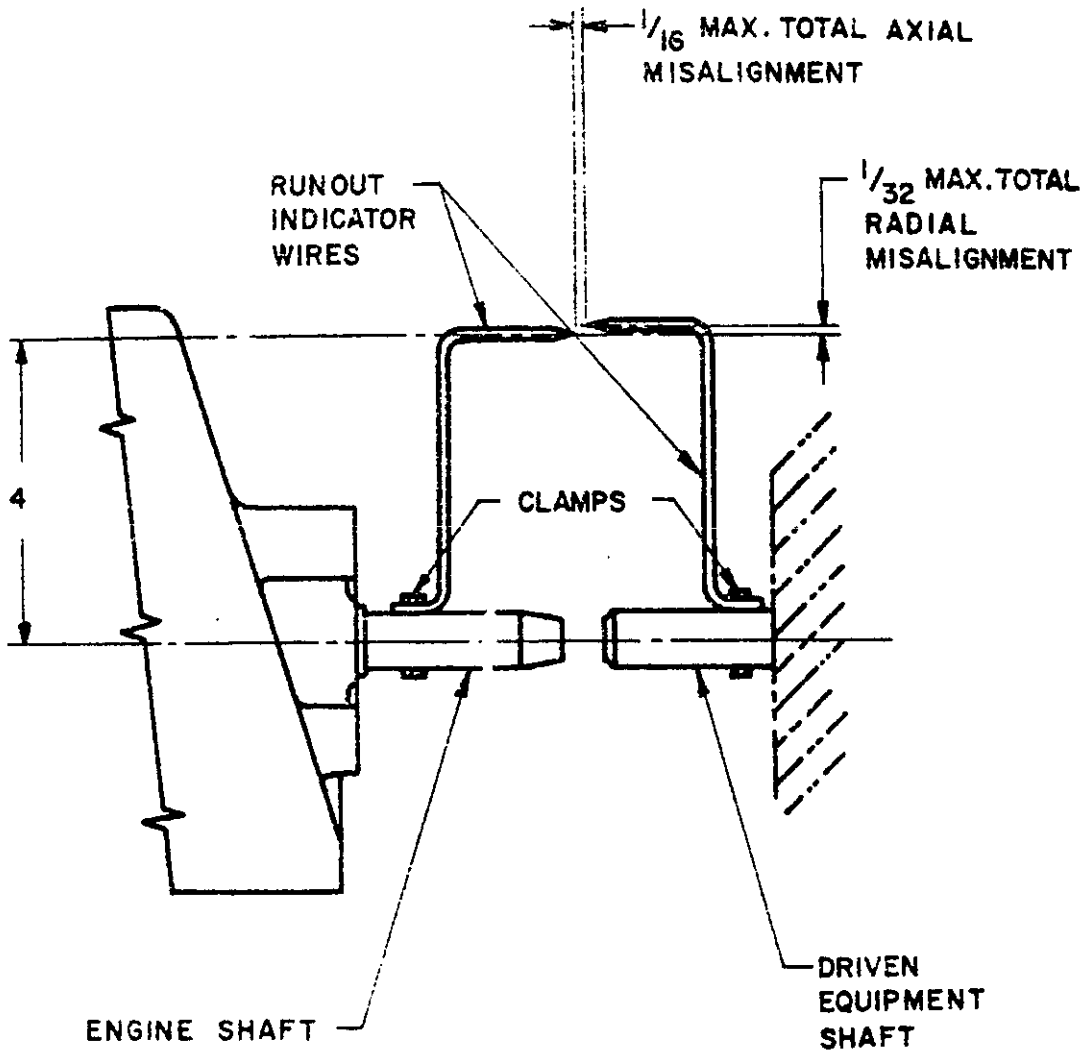



FIGURE 11. *Alignment method.*

CX–427



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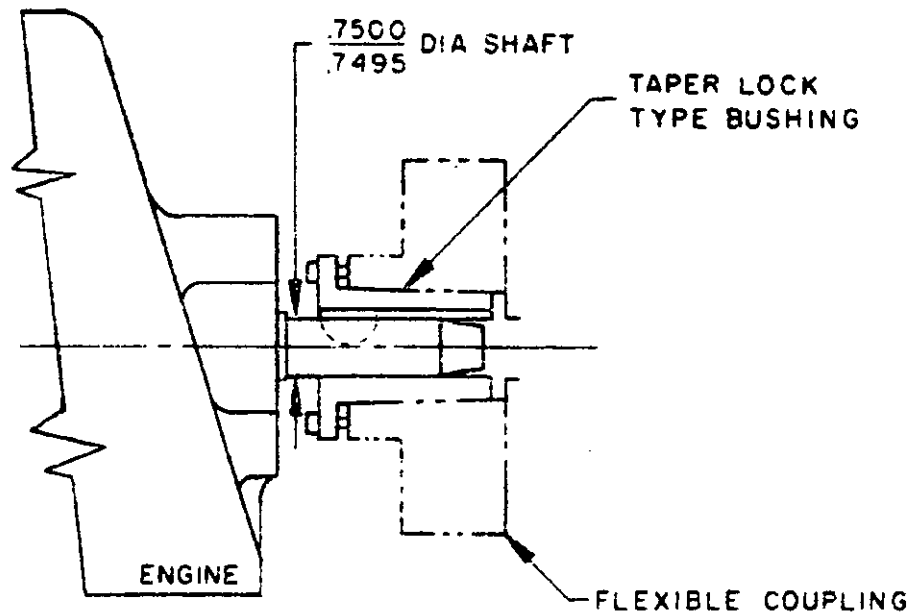



FIGURE 12. *Flexible coupling installation.*

CX-429



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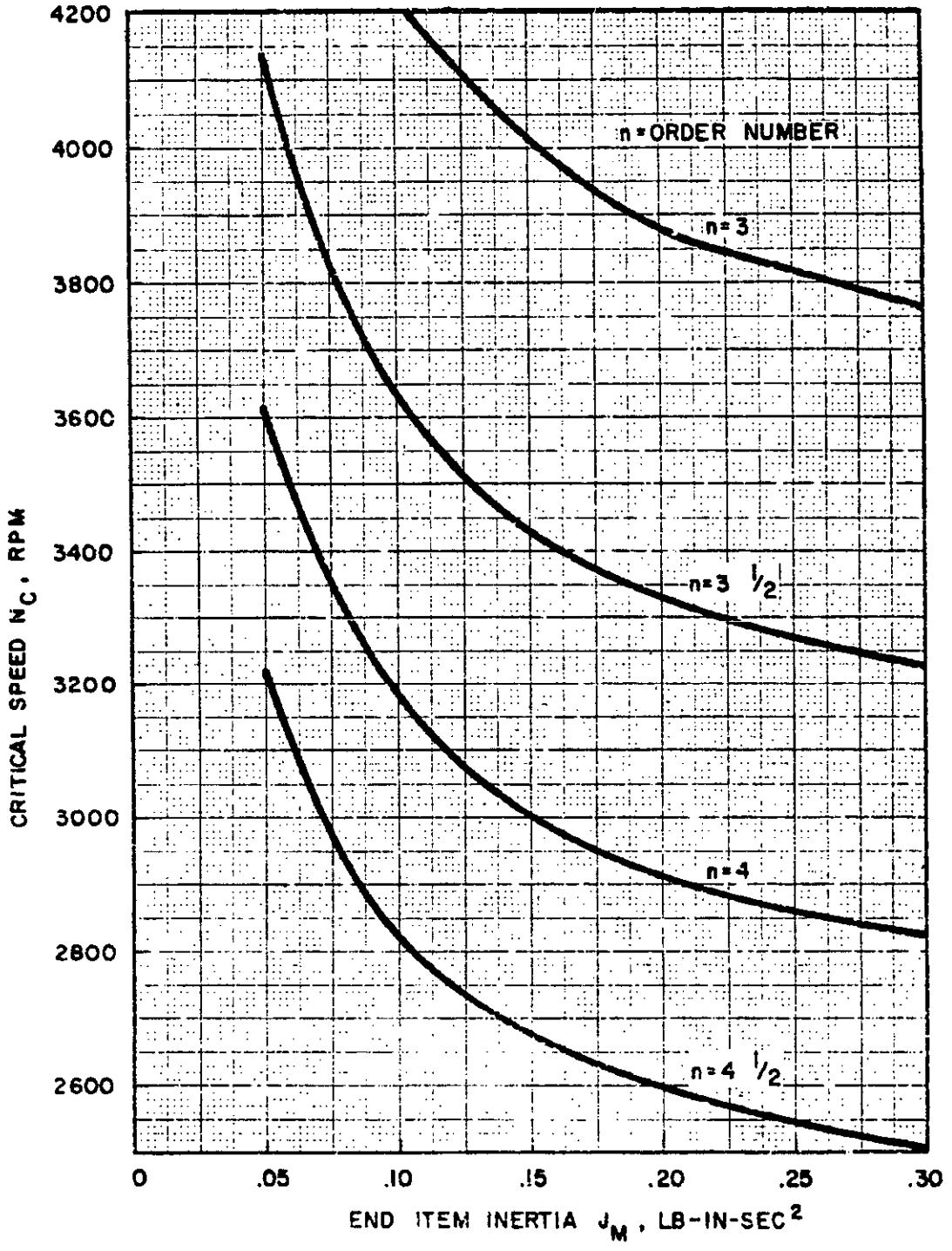



FIGURE 13. *Critical speeds for directly attached end item.*

CX-442

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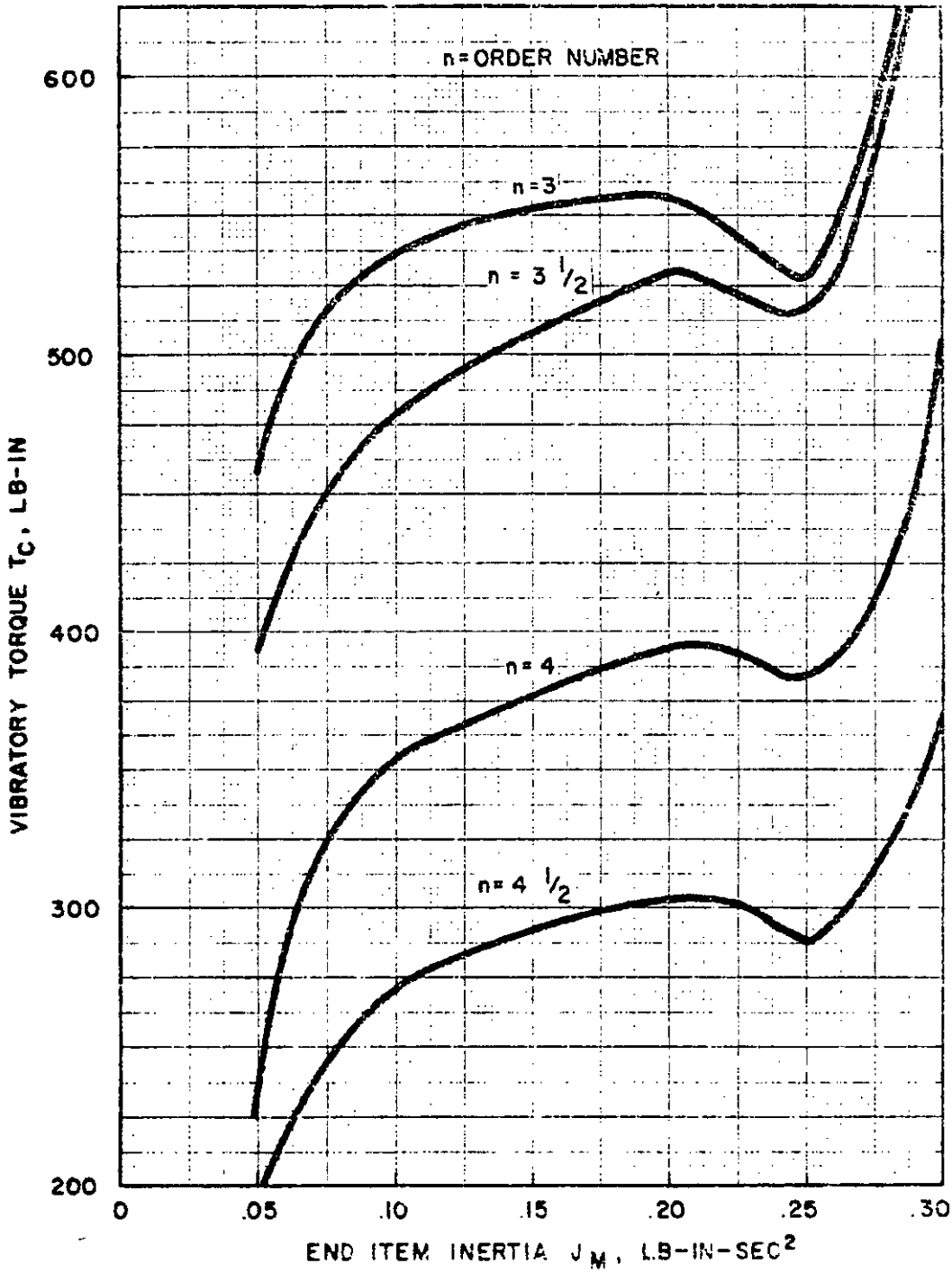



FIGURE 11. *Vibratory torque at critical speeds for directly attached end item.*

CX-443



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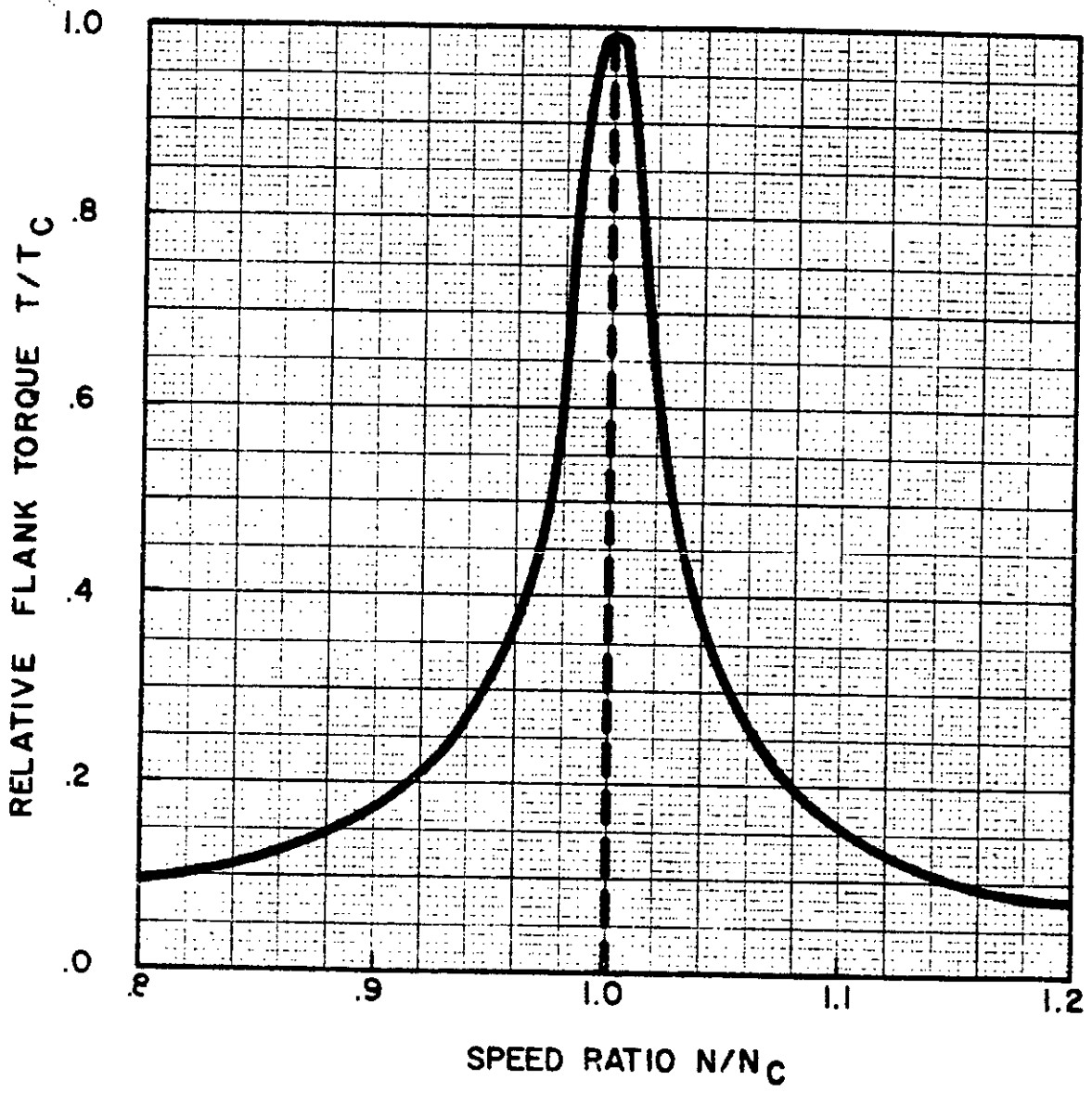



FIGURE 15. *Average torque magnifier for directly attached end item.*

CX–444

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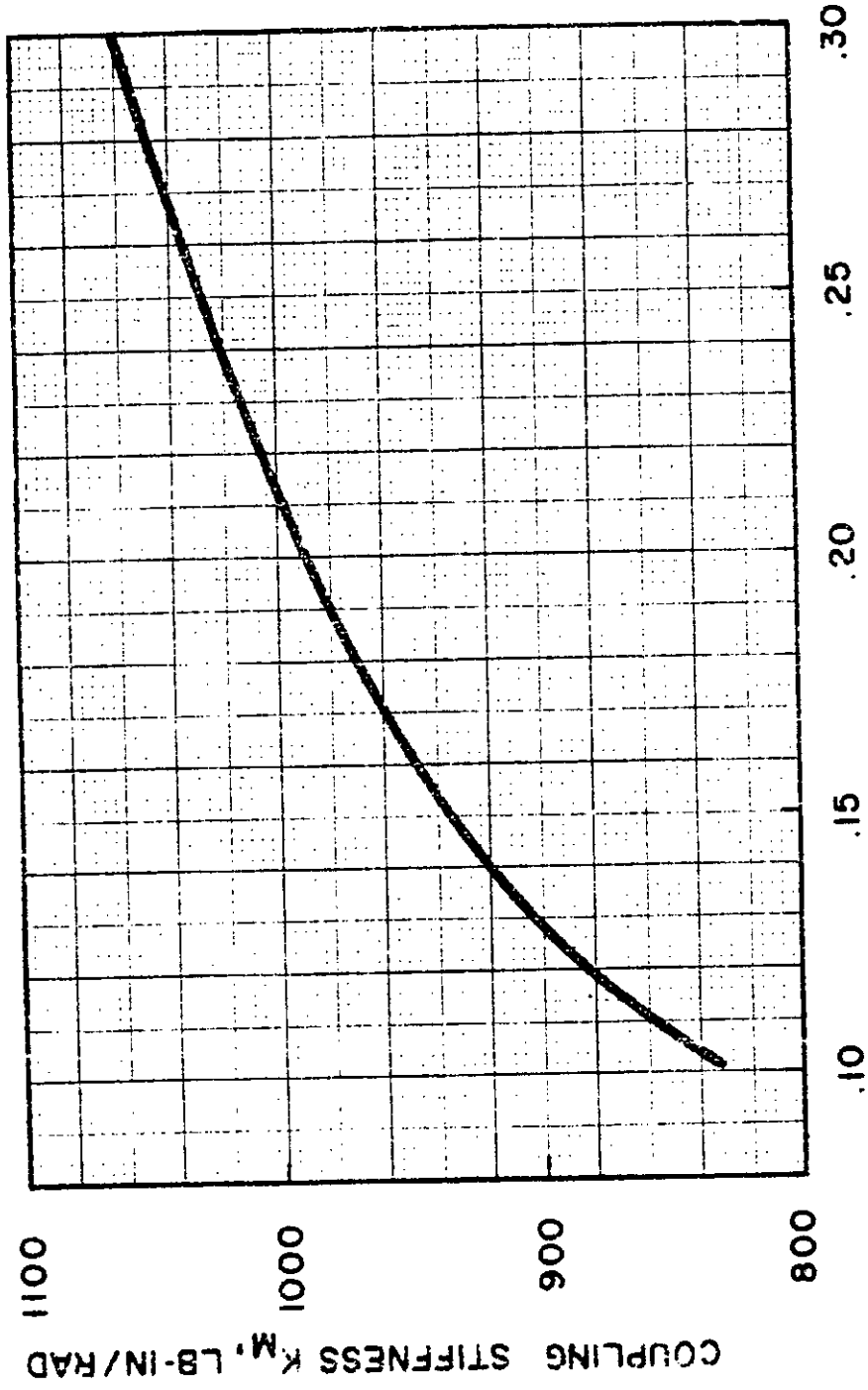



FIGURE 16. *Permissible maximum stiffness of flexible coupling*



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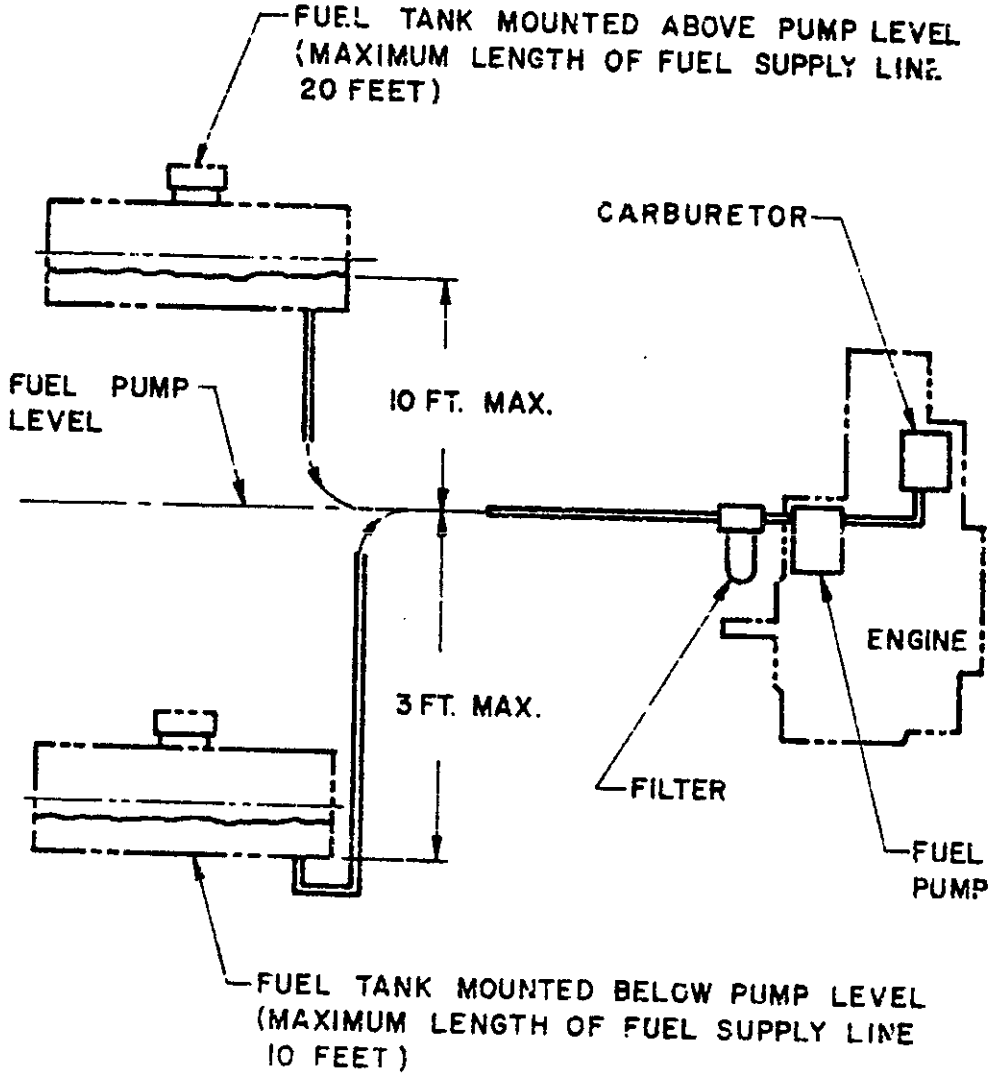



FIGURE 17. *Maximum limits for fuel tank locations.*

CX-428

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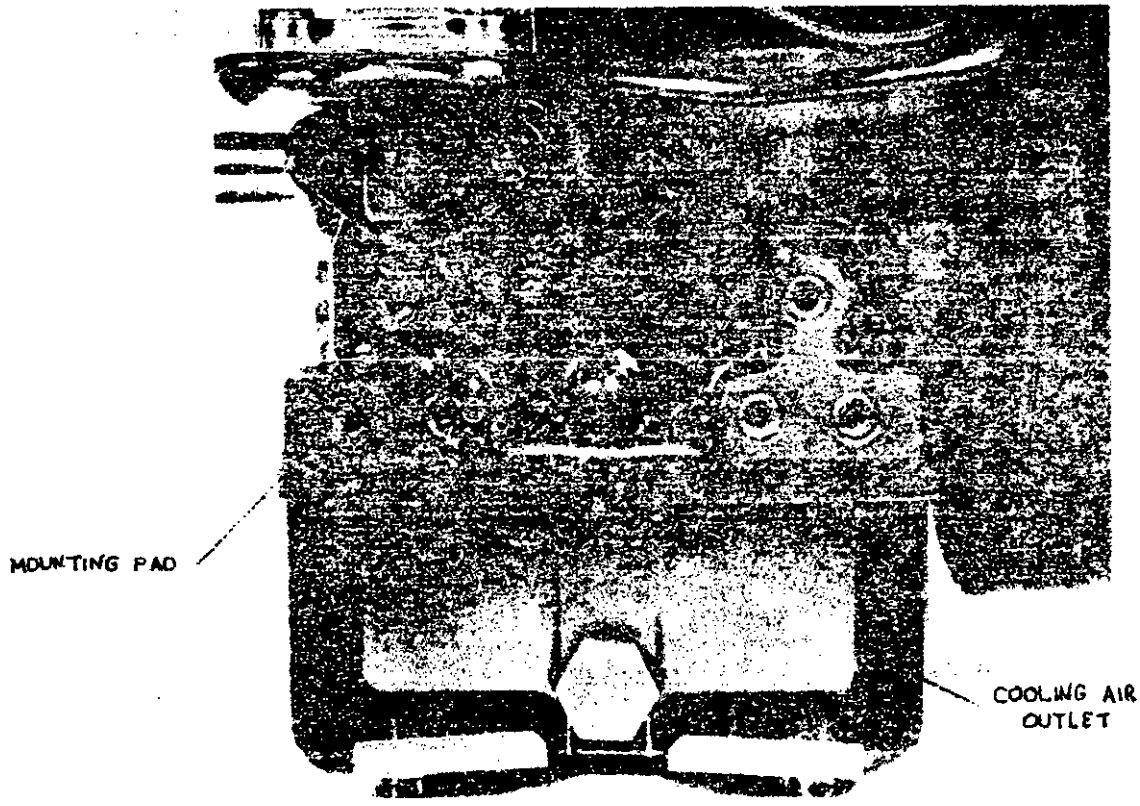



FIGURE 18. *Cooling air outlet and engine mounting pads model J-108 military design engine.*



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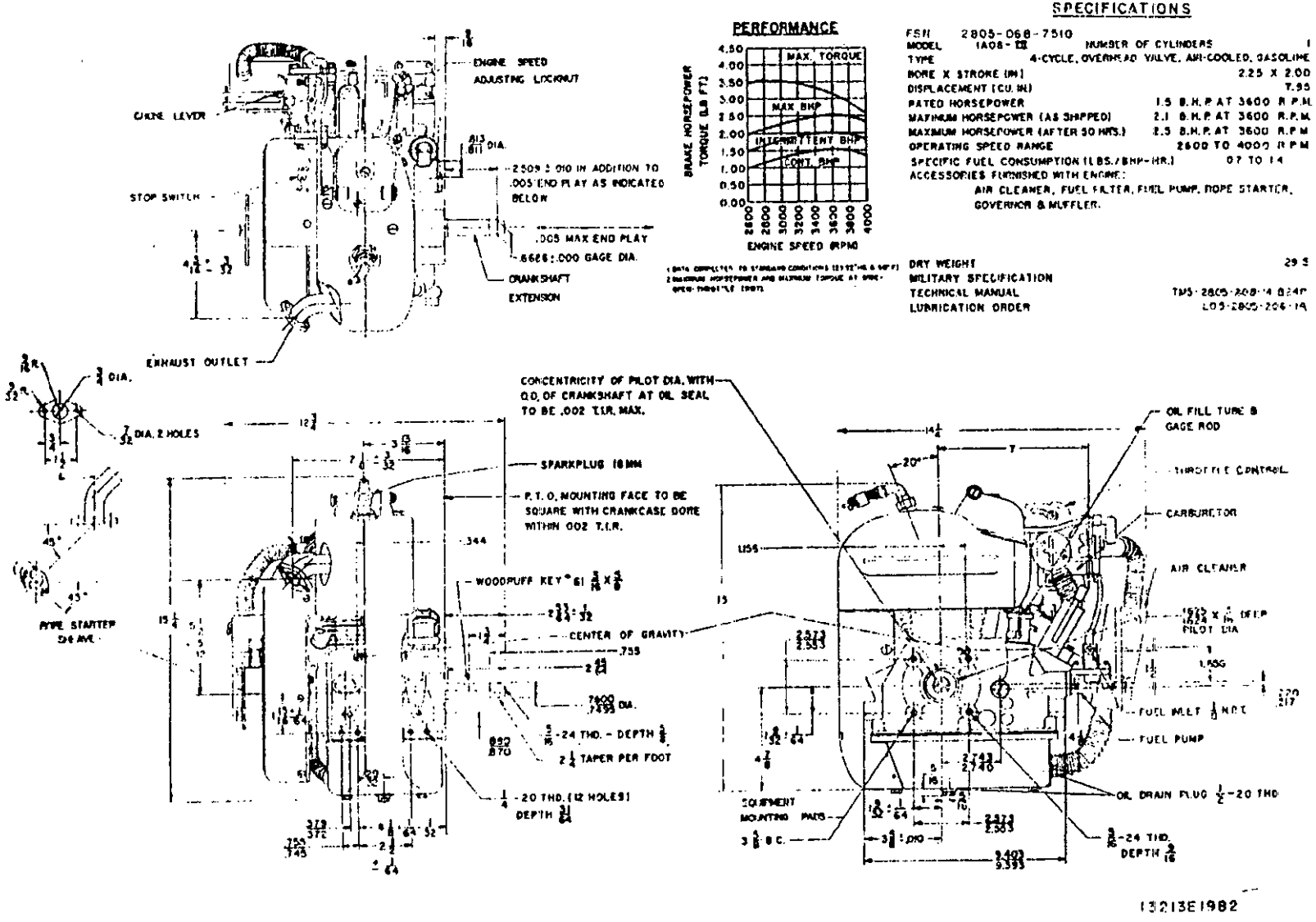
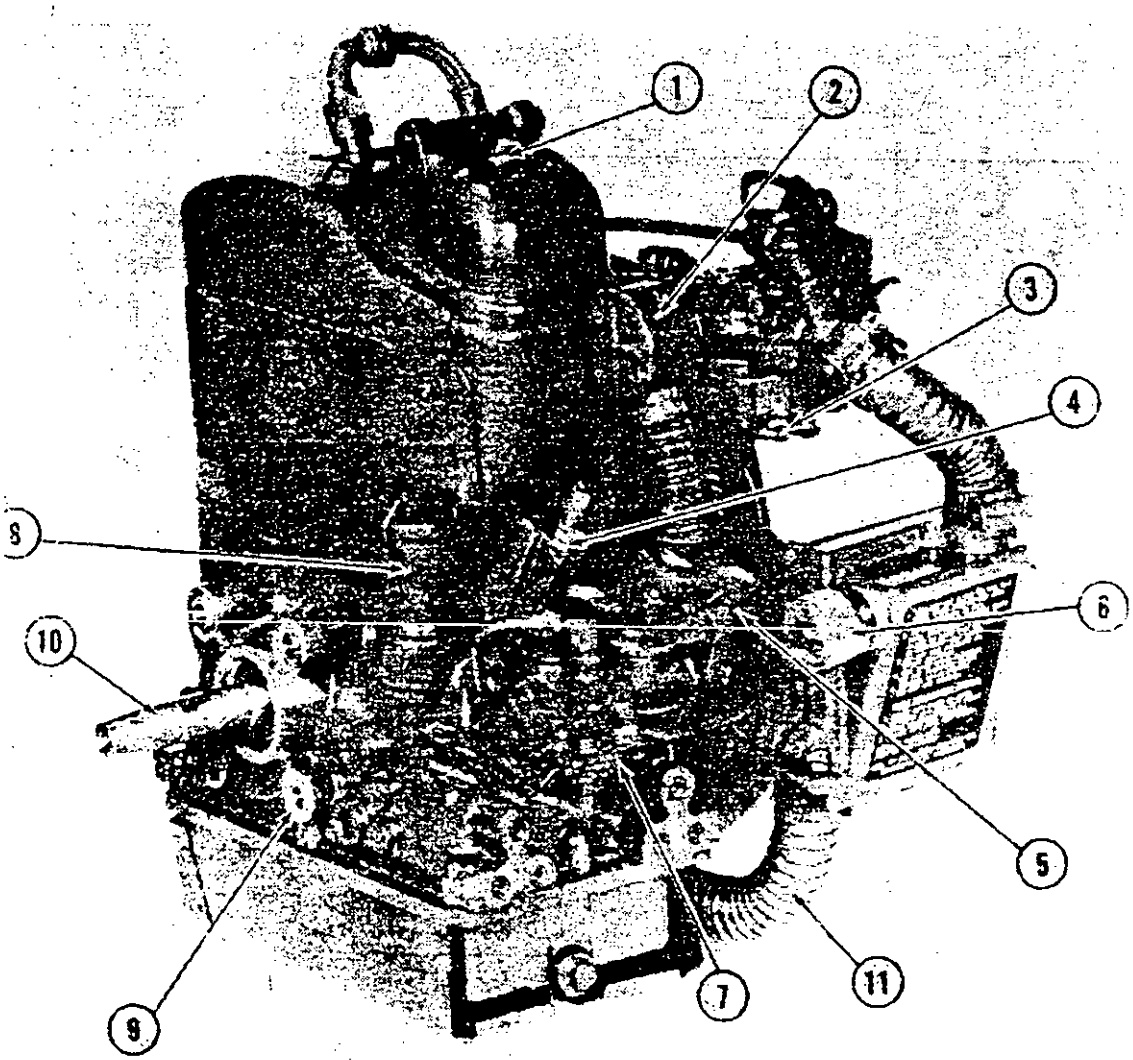



FIGURE 19. *Model 1A08-III 1 ½ HP MS engine, installation drawing.*





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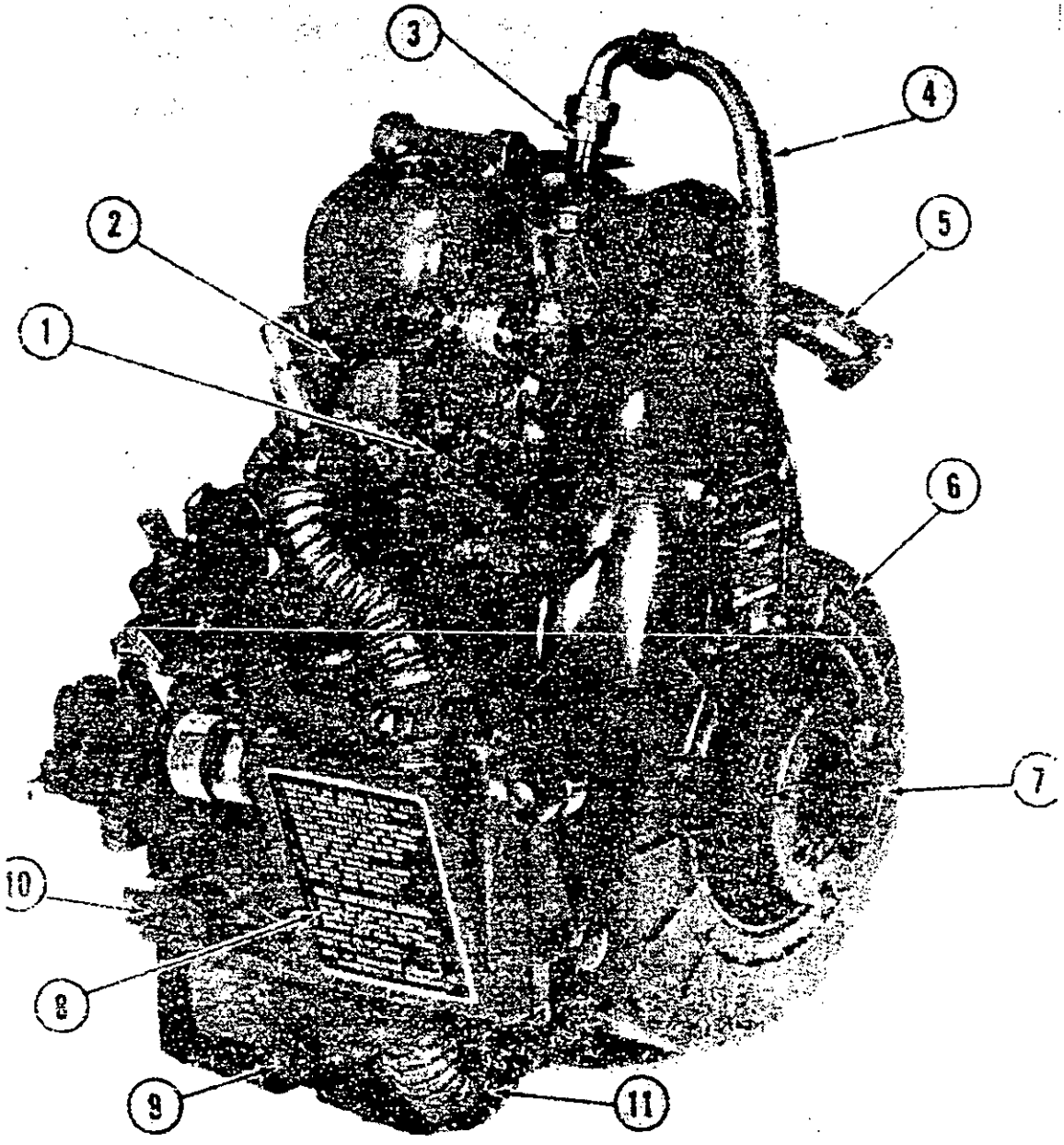



1. Rocker Box Cover
2. Throttle Housing
3. High Speed Mixture Needle
4. Engine Speed Adjusting LocKnut
5. Distributor Cover
6. Air Cleaner Restriction indicator
7. Fuel Pump
8. Oil Filter Tube and gauge Rod
9. Equipment Mounting Pads
10. Power Take-off Shaft
11. Warm Air Duct

FIGURE 29. *Model TA08-III Military design engine, power take-off end.*



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1. Choke Lever
2. Carburetor
3. Spark Plug
4. High Tension Cable
5. Exhaust Outlet
6. Flywheel Fan Housing
7. Rope Starter Pulley
8. Dry Air Cleaner
9. Oil Drain Plug
10. Engine Mounting Pads
11. Warm Air Duct

FIGURE 21. *Model LA08-III Military design engine rope starter end.*

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